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How Peer Discussion of Cervical Cancer Screening Centers on the Exam: Topic Modeling and Network Analysis Study

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Abstract

Background: Cervical cancer is largely preventable, yet it caused about 604,000 new cases and 280,000 deaths worldwide in 2024. Fear can motivate or deter screening, and fear of cancer differs from fear of the exam itself. Online health communities record these concerns in users' own words, which makes them a practical source of insight for screening programs.

Objective: This study mapped the topics of cervical cancer screening discussion in peer online health communities, how those topics connect, and where negative emotion and fear concentrate.

Methods: We analyzed 9851 posts and comments from 7 women's health subreddits posted between August 1, 2025, and January 31, 2026. Latent Dirichlet allocation and the Analysis of Topic Model Networks procedure identified topics, clusters, and bridging topics. A transformer sentiment model classified each document as negative, neutral, or positive; we validated it, a second transformer model, and the VADER (Valence Aware Dictionary and sEntiment Reasoner) lexicon against 2 coders' consensus coding of 209 documents. We counted explicit fear terms by topic and their co-occurrence with procedure and disease terms.

Results: The model yielded 13 substantive topics that formed 3 clusters (modularity 0.30): diagnostic information, the embodied experience of the exam, and personal narrative with peer support. Timing and bleeding, the least prevalent topic (3.5%), had the highest betweenness centrality (0.26) across network thresholds. The transformer model agreed with human coding more closely (macro F1=0.61) than VADER (macro F1=0.41), which labeled 40% of human-coded negative documents as positive. By the transformer model, 36.8% of documents were negative and 19.6% positive, and encouragement and support was the only topic with more positive than negative documents. Fear terms appeared in 3.5% of documents on human papillomavirus (HPV) and cancer risk, the lowest share of any topic, against 21.8% for pain and discomfort and 27.0% for accounts of clinical visits. Sentences expressing fear mentioned the procedure without the disease far more often than the reverse (559 vs 66). HPV and cancer risk connected to the exam-experience cluster only through access to care.

Conclusions: In this discourse, fear attaches to the screening exam far more than to cervical cancer, and the topics that connect the conversation concern bleeding, pain, and getting an appointment. Screening communication may gain more from building confidence about the exam, through clear information on what to expect, pain management, and self-sampling options, than from further emphasis on cancer risk. Lexicon-based sentiment tools should be used with caution in health contexts, where words such as positive carry clinical meaning.

Keywords: cervical cancer screening; Papanicolaou test; online health communities; Reddit; topic modeling; network analysis; fear appeals; infodemiology; social listening; health communication

Introduction

Background

Cervical cancer is largely preventable through human papillomavirus (HPV) vaccination and regular screening. About 604,000 women received a diagnosis and about 280,000 died in 2024, and incidence and mortality are highest in low- and middle-income countries [1]. Treating the precancerous lesions that screening detects is a simple and effective way to prevent the disease [1]. The World Health Organization therefore asks countries to vaccinate 90% of girls by age 15, screen 70% of women with a high-performance test by age 35 and again by age 45, and treat 90% of women with precancer or cancer [1].

Screening coverage falls short of that target in many settings, and the shortfall has psychosocial as well as health-system causes [2]. Many of the psychosocial barriers concern the exam itself. Women describe pain, embarrassment, uncertainty about what the exam involves, and negative experiences with clinicians as barriers to screening [3,4]. These barriers have recently become easier to address. In 2024, the US Food and Drug Administration approved HPV tests that allow self-collection for people who cannot have or do not want a pelvic exam [5], and self-sampling is highly acceptable to women [6]. Whether such options, and the messages that promote screening, reach the people who stay away depends on knowing which concerns dominate, which ones carry fear, and how those concerns connect to one another.

People increasingly voice those concerns online. Social media use has spread widely, with implications for health communication [7], and pseudonymous, threaded platforms such as Reddit encourage disclosure that people withhold in person [8,9]. For screening programs, this discourse is a standing,

unprompted record of what the target population worries about. Social listening turns that record into evidence that can guide message design.

Prior Work

Computational studies have profiled health talk on social media at scale. Researchers have used topic modeling to map obesity discussion on Twitter [10] and combined multiple Twitter data collection tools to track HPV vaccine communication [11]. A systematic review documents a growing body of health communication research on Reddit [9], and recent work applies computational grounded approaches to risk talk in online communities [12]. Many such studies also score sentiment with general-purpose tools such as VADER [13,14]. An evaluation on health-related social media data found that such tools, developed outside health care, classified sentiment poorly [15], and sentiment analysis of medical text has been argued to require domain-specific sentiment resources [16].

Most of this work reports which topics appear and how often. Less attention has gone to how topics connect. Two communities can discuss the same topics and still organize them differently, and that organization matters for practice. A topic that sits between otherwise separate parts of a conversation reaches audiences that a peripheral topic cannot. Network agenda setting offers a vocabulary for this structure, treating an agenda as a set of associations among topics [17,18]. The present study borrows that analytic lens and applies it to a peer community, where no editor or institution sets the agenda.

Fear and Screening

Fear enters screening decisions in 2 forms, and research on fear and screening has struggled to separate them. Fear appeal theory holds that a perceived threat motivates protective action when people believe the action is effective and within reach, and that strong threat combined with low efficacy produces defensive responses [19,20]. Messages that stress the severity of cervical cancer and personal susceptibility work on the threat side of this model. In screening, however, the protective action can itself be feared. Women describe fear of pain, embarrassment, and the speculum exam as barriers separate from fear of what the test might find [3,4,21]. Studies of cancer fear have linked it to higher screening uptake in some samples

and lower uptake in others [22], and a review of breast cancer screening called for research that separates the aspects of cancer and of the screening process that women fear, because each may relate differently to screening behavior [23]. Much of this evidence comes from surveys and interviews, in which the researcher's questions set the topic. Less is known about which fear people raise on their own when they discuss screening with peers, and where that fear sits in the structure of their discussion. The answer bears on whether screening messages should raise threat or build efficacy. If the fear that keeps people from screening concerns the exam, messages that heighten the threat of cancer without addressing the exam may create the combination of strong threat and low efficacy under which fear appeals produce defensive responses [19,20].

Objective

This study examined cervical cancer screening discourse in peer online health communities from 2 angles that bear on message design. The topic network shows where concerns sit and which ones connect otherwise separate parts of the conversation, which informs where information should be placed. The emotional content of each topic, and fear in particular, shows which concerns carry distress and what that distress is about, which informs what messages should say. Four questions guided the analysis:

RQ1: Which topics make up cervical cancer screening discourse on Reddit, and how prevalent is each?

RQ2: How do these topics connect, and what clusters do they form?

RQ3: Which topics bridge the clusters?

RQ4: How do sentiment and fear-related language vary across topics, and what do expressions of fear refer to?

The study makes 3 contributions. First, it shows what people fear when they discuss screening unprompted, a question an earlier review raised for breast cancer screening [23]. Second, it locates where concerns connect in the conversation, which identifies where screening information can reach the most people. Third, it shows how much the choice of sentiment tool changes conclusions about health discourse, a practical concern for social listening.

Methods

Study Design

This was a cross-sectional computational content analysis of publicly available Reddit posts and comments. The analysis proceeded in five stages: corpus construction, topic modeling, topic-network construction, sentiment classification with human validation, and analysis of fear-related language.

Data Collection

We collected data through the official Reddit application programming interface. The sampling frame comprised 7 subreddits in which cervical screening is discussed regularly: r/HPV, r/WomensHealth, r/PreCervicalCancer, r/obgyn, r/CervicalCancer, r/Healthyhooha, and r/vaginismus. We retained submissions posted between August 1, 2025, and January 31, 2026, whose title or body matched at least one screening-related keyword: Pap smear, Pap test, Pap, smear test, HPV test, cervical screening, or gyno exam. Matching on the term Pap required a word boundary to avoid false matches. We then retrieved all comments on every matched submission, without applying the keyword filter to comments, so that replies remained attached to the discussions that prompted them.

The search returned 1105 submissions and 9413 comments. We removed content deleted or removed by users or moderators and documents shorter than 30 characters, which left 9851 documents: 1104 submissions and 8747 comments. Reddit's search returns a ranked set of matches and does not retrieve every match, so the corpus is a large sample of this discourse.

We treated submissions and comments as documents in a single corpus for three reasons. First, screening discussion on Reddit unfolds as exchanges: a submission asks a question or describes an experience, and comments supply information, advice, and reassurance. Second, comments made up 89% of the corpus (8747 of 9851 documents), so a submissions-only corpus would omit most of the community's responses, including the peer support that the topic network is meant to locate. Third, every comment came from a thread whose submission matched a screening keyword, so comments shared the screening context of the discussion. Individual submissions and comments were the most common unit of analysis in a review of

Reddit research on substance use [24], and topic-modeling studies of Reddit health discussion have combined them in a single corpus [25,26]. Comments are shorter than submissions and are nested within threads, a point we return to in the Limitations.

Preprocessing

We lowercased each document, stripped URLs and non-alphabetic characters, and tokenized the text. We removed English stopwords and a custom list of platform and URL fragments (eg, reddit, removed, deleted), contraction fragments (eg, dont, cant), and conversational filler (eg, just, like, thanks), and kept tokens of at least 3 letters. We then built a document-term matrix of unigrams and bigrams appearing in at least 10 documents and no more than 40% of documents, capped at 4000 terms. Twenty-three documents contained no in-vocabulary term after this step and dropped out, leaving 9828 documents for modeling.

Topic Modeling

We fit latent Dirichlet allocation (LDA) [27], which assigns every document a probability distribution across all topics. The network stage requires that full distribution. Clustering-based models such as BERTopic assume that each document contains a single topic, and their document-topic probabilities serve only as an approximate proxy for a document's topic mixture [28]. Applications of the Analysis of Topic Model Networks (ANTMN) procedure in health communication have accordingly used LDA [29-31]. We selected the number of topics by fitting models with k from 8 to 24 in steps of 2 and comparing UMass coherence. Coherence was highest at small k, reached a local optimum at k=14 (coherence -1.55), and declined steadily above it. We retained k=14 because it maximized coherence among solutions granular enough to support a topic network. Models in the scan used online variational Bayes; the final model used batch variational Bayes with 60 iterations. One topic consisted entirely of automoderator and community-rule text and was excluded as non-substantive, leaving 13 topics. We did not remove individual automoderator comments before modeling. LDA isolated their language in this topic, which was the highest-loading topic for 159 documents. Analyses that assign documents to topics used each document's highest-loading substantive topic. The author read the 20 highest-loading documents for each topic and assigned a label; all 13 corresponded to a coherent and distinct theme.

Topic Network Construction

We followed the ANTMN procedure [32,33], using the Spearman-similarity and disparity-filter specification of Liu et al [34] in place of the cosine similarity of the original procedure. For each pair of topics we computed the Spearman rank correlation between their loading columns across all documents. Spearman correlation captures monotonic co-occurrence and tolerates the skewed distribution of topic prevalence. We set negative correlations to zero because the analysis concerns co-occurrence and the backbone algorithm requires non-negative weights.

A fully connected correlation network is dense and noisy. We therefore extracted its statistically significant backbone with the disparity filter [35], which retains edges whose weight is significantly larger than expected if a node's strength were distributed uniformly at random across its edges. We used a significance level of alpha=.30, the value that produced a single connected and interpretable backbone for a network of this size. The Louvain algorithm [36] detected clusters on the weighted backbone. Node prevalence equals the mean of each topic's loading column.

We characterized the structural position of each topic with strength (the sum of its edge weights), degree, betweenness centrality, closeness centrality, eigenvector centrality, and Burt constraint [37]. Because edge weights measure similarity, betweenness and closeness used inverse weights as distances. Betweenness identifies topics that lie on the paths connecting otherwise separate clusters.

Sentiment Analysis

We classified each document as negative, neutral, or positive with twitter-roberta-base-sentiment-latest, a RoBERTa-base transformer model trained on about 124 million tweets and fine-tuned for 3-class sentiment on the TweetEval benchmark [38,39]. Each document received the label with the highest predicted probability. We summarized sentiment as the share of negative, neutral, and positive documents in each topic and cluster, assigning each document to its highest-loading substantive topic.

We chose the model after validating it against human coding. Lexicon tools assign each word a fixed valence, and clinical language reuses everyday words with different meanings. In the lexicon of VADER,

a rule-based tool developed for social media text [13], positive carries a valence of +2.6 and negative a valence of -2.7 [40], so a report of an HPV-positive result can register as positive emotion. We compared 3 tools against human judgment: twitter-roberta-base-sentiment-latest; VADER, with compound scores classified by the typical thresholds given in its documentation (≥0.05 positive, ≤-0.05 negative) [40]; and distilbert-base-uncased-finetuned-sst-2-english, a 2-class transformer model fine-tuned on movie-review sentences [41,42].

The validation sample contained 209 documents in 2 strata. Stratum A comprised 130 documents drawn at random from the 13 topics (10 per topic). Stratum B comprised 79 documents drawn at random from documents containing explicit test-result language (eg, tested positive, came back negative, abnormal), where clinical and emotional meaning are most likely to diverge. Two coders, the author and a second coder, independently coded the emotion each writer expressed on a 5-point scale from -2 (strongly negative) to +2 (strongly positive) and disregarded the clinical meaning of any result. Both were blind to all automated scores and to each other's codes. Coding used the English text; one Spanish-language document was translated into English before coding. After two rounds of independent coding, inter-coder reliability was Krippendorff $\alpha=0.76$ [43]. The coders resolved disagreements in the direction of sentiment by discussion. Where they agreed on direction and differed only in intensity, we used the mean of their 2 codes. The resulting consensus served as the reference standard. Three sampled documents were automoderator or moderation messages; we retained them to match the analyzed corpus and coded them like other documents.

We prespecified the 3-class comparison (negative, neutral, positive) as the primary analysis because the direction of expressed emotion can be judged more reliably than its intensity, and because the transformer model outputs 3 classes. For each tool we report accuracy and macro-averaged F1 against the reference standard. As a secondary analysis, we report the Spearman correlation between the 5-point reference code and each tool's continuous score, with a paired bootstrap (5000 resamples) for the difference between the transformer model and VADER.

Fear-Related Language

To examine fear specifically, we counted explicit fear and anxiety terms (eg, afraid, scared, terrified, anxious, nervous, panic, freaking out). We restricted the list to explicit emotion words so that the count would not depend on clinical vocabulary, the problem the validation documented for VADER. For each topic we report the share of documents containing at least one such term. Because LDA assigns every document a mixture of topics, we also compared the mean topic loadings of documents with and without a fear term, which shows which topics fear-bearing documents draw on. To see what the fear concerned, we split documents into sentences and recorded whether each sentence containing a fear term also mentioned the screening procedure or exam (eg, Pap, speculum, colposcopy, biopsy, pain), a test result (eg, result, abnormal, positive), or the disease (eg, cancer, death). Multimedia Appendix 1 lists all terms. As robustness checks, we repeated the analysis with worry terms added and with the fear category of the NRC Emotion Lexicon [44], a crowdsourced word-emotion lexicon. We ran the NRC check in 2 versions: all NRC fear words, and NRC fear words minus terms that name a disease, symptom, body event, or medical procedure (eg, cancer, pain, biopsy), because such terms mark the topic of a post as much as the writer's emotion. In the sentence-level analysis, procedure and disease terms never counted as fear terms. The analysis is exploratory: co-occurrence within a sentence shows what writers mention alongside fear, which approximates the object of the fear without establishing it.

Statistical Analysis

Analyses ran in Python 3.11 using scikit-learn for LDA, vectorization, and agreement statistics, SciPy for Spearman correlations, NetworkX and python-louvain for the network analysis, and Hugging Face transformers and vaderSentiment for sentiment. All random processes used a fixed seed. To test whether the findings depend on the network threshold, we repeated the backbone extraction, clustering, and centrality analysis across alpha values from .25 to .35 and compared cluster membership and the rank order of betweenness.

Ethical Considerations

This study analyzed publicly available content from public subreddits. It involved no interaction or intervention with individuals and collected no identifiable private information, so it does not meet the federal definition of human subjects research. The University of California, Santa Barbara Human Subjects Committee reviewed the project and determined on October 8, 2026, that it does not meet the definition of human subjects research under the Common Rule (45 CFR 46.102) and does not require institutional review board review. Account handles in the platform export were discarded during preprocessing and were never analyzed, linked, or reported.

Informed consent cannot be obtained for an analysis of archived public posts, so we evaluated the study against considerations proposed by Eysenbach and Till [45]. The first concerns whether members perceive the space as public or private. All 7 subreddits are open, require no approval to join or read, and are indexed by search engines, which places them at the public end of that continuum. The second concerns intrusiveness. The analysis was non-intrusive: we did not post, recruit, contact users, or take part in any discussion. The third concerns the risk that reporting identifies an individual. Verbatim quotations from public posts can be traced back to an account through a search engine, so we paraphrase every illustrative example and report no usernames, dates, or thread identifiers. Because the corpus contains personal health disclosures, we do not redistribute it.

Results

Topics in Screening Discourse (RQ1)

The 13 substantive topics covered the screening pathway from routine testing and result interpretation through follow-up procedures, symptoms, pain, access to care, and peer support (Table 1). The most prevalent topics were encouragement and support (11.2%), pain and discomfort (10.9%), doctor distrust and communication (10.1%), colposcopy and loop electrosurgical excision procedure (LEEP) follow-up (9.8%), and personal accounts of clinical visits (8.4%). The least prevalent was timing and bleeding (3.5%).

Table 1. Topics in cervical cancer screening discourse on Reddit, with representative terms, prevalence, sentiment, and cluster (N=9828 documents).

Topic	Representative terms	Prevalence, %	Negative / positive, %	Fear terms, %	Cluster
Encouragement and support	hope, good, best, feel, help, support	11.2	22 / 54	12.3	Narrative and support
Pain and discomfort	pain, painful, experience, speculum	10.9	49 / 24	21.8	Exam experience
Doctor distrust and communication	people, say, tell, doctor, health	10.1	41 / 16	9.1	Diagnostic information
Colposcopy or LEEP follow-up	leep, cin, biopsy, colposcopy, cells	9.8	27 / 17	16.7	Diagnostic information
Accounts of clinical visits	said, told, went, weeks, procedure	8.4	43 / 19	27.0	Narrative and support
Test results (HPV or Pap)	hpv, pap, positive, negative, abnormal	8.3	31 / 11	18.0	Diagnostic information
HPV transmission and warts	warts, infection, immune, partner	7.2	25 / 9	4.2	Diagnostic information
Access to care and exams	doctor, medical, exam, appointment	7.2	39 / 19	5.1	Exam experience
Routine Pap testing	pap smear, smears, sexually active	6.9	46 / 7	14.3	Exam experience
Symptoms and infections	yeast, vaginal, discharge, birth control	5.9	49 / 7	6.2	Narrative and support
Pelvic pain and vaginismus	pelvic, vaginismus, floor, penetration	5.7	43 / 18	15.8	Exam experience
HPV and cancer risk	cancer, cervical, risk, screening	5.1	41 / 5	3.5	Diagnostic information
Timing and bleeding	period, months, days, bleeding	3.5	32 / 10	16.9	Narrative and support

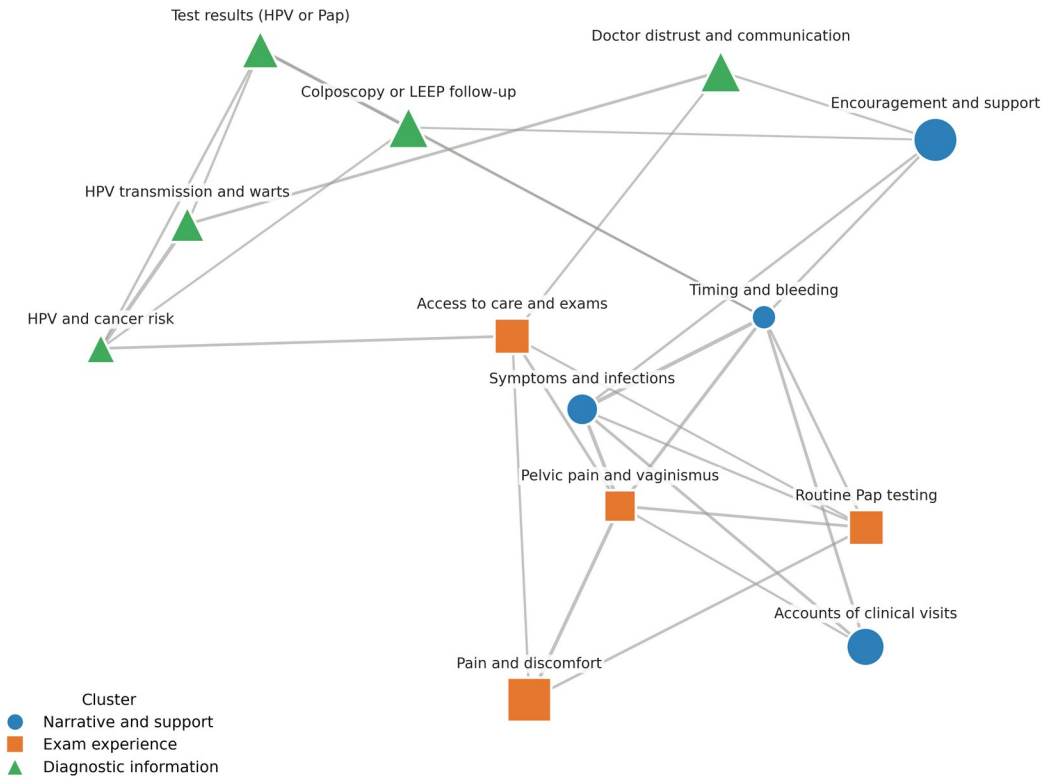
Prevalence is the mean document-topic loading. Negative / positive is the share of documents classified as negative and as positive by twitter-roberta-base-sentiment-latest among documents whose highest-loading substantive topic is the row topic; the remainder were classified as neutral. Fear terms is the share of documents containing at least one explicit fear or anxiety term (Multimedia Appendix 1). Cluster is the Louvain assignment on the disparity-filter backbone shown in Figure 1. HPV: human papillomavirus; LEEP: loop electrosurgical excision procedure.

Network Structure and Clusters (RQ2)

The backbone network contained 13 nodes and 28 edges, with a density of 0.36, and formed a single connected component. Louvain detected 3 clusters with a modularity of 0.30 (Figure 1). The first cluster covered diagnostic information: test results, HPV and cancer risk, HPV transmission, colposcopy or LEEP follow-up, and doctor communication. The second covered the embodied experience of the exam: routine Pap testing, pain and discomfort, pelvic pain and vaginismus, and access to care. The third covered personal narrative and support: accounts of clinical visits, symptoms, timing and bleeding, and

peer encouragement. By prevalence, the diagnostic information cluster accounted for 40.5% of the discourse, the exam-experience cluster for 30.7%, and the narrative and support cluster for 29.0%.

Figure 1. Topic network for cervical cancer screening discourse on Reddit.



Nodes are topics. Node size is prevalence, and node color and shape indicate Louvain cluster. Edges are the disparity-filter backbone (alpha=.30) of the Spearman topic-loading similarity network, and edge width is proportional to similarity.

Bridging Topics (RQ3)

Topics differed in how far they reached across the network (Table 2). Timing and bleeding had the highest betweenness centrality (0.26) together with the highest strength (2.28) and degree (0.58). Pelvic pain and vaginismus and access to care followed, each with a betweenness of 0.15. HPV and cancer risk had the highest betweenness among the diagnostic topics (0.11). Doctor distrust and communication and HPV transmission and warts had low betweenness (0.03 each). Accounts of clinical visits and pain and discomfort had the highest Burt constraint (0.66 and 0.63), indicating comparatively closed neighborhoods in which their few connections ran to topics that were themselves connected.

Table 2. Network position of each topic on the weighted disparity-filter backbone.

Topic	Strength	Degree	Betweenness	Closeness	Eigenvector	Constraint
Timing and bleeding	2.28	0.58	0.26	0.22	0.45	0.36
Pelvic pain and vaginismus	2.08	0.50	0.15	0.21	0.45	0.43
Access to care and exams	1.42	0.42	0.15	0.19	0.24	0.35
HPV and cancer risk	1.29	0.33	0.11	0.17	0.13	0.42
Test results (HPV or Pap)	1.20	0.33	0.08	0.18	0.16	0.49
Encouragement and support	1.07	0.33	0.06	0.17	0.21	0.37
Symptoms and infections	1.74	0.42	0.03	0.19	0.41	0.49
Doctor distrust and communication	0.86	0.25	0.03	0.15	0.10	0.34
Colposcopy or LEEP follow-up	1.10	0.33	0.03	0.16	0.17	0.45
HPV transmission and warts	1.04	0.25	0.03	0.15	0.09	0.46
Pain and discomfort	0.97	0.25	0.00	0.16	0.23	0.63
Routine Pap testing	1.46	0.42	0.00	0.18	0.34	0.49
Accounts of clinical visits	0.94	0.25	0.00	0.15	0.27	0.66

Strength is the sum of incident edge weights. Betweenness and closeness use inverse edge weights as distances. Constraint is the Burt measure, where higher values indicate a more closed neighborhood. Rows are ordered by betweenness. HPV: human papillomavirus; LEEP: loop electrosurgical excision procedure.

Reading the highest-loading documents for the three bridging topics showed what connects the clusters. In the timing and bleeding topic, users who had a Pap test days or weeks earlier asked whether a delay in results or unexpected bleeding signaled a problem. In the pelvic pain topic, users urged one another to request a referral to pelvic floor physical therapy for suspected vaginismus. In the access topic, users debated whether a routine gynecological exam was necessary and how to learn in advance what it would involve. (Examples are paraphrased.)

Sentiment Classification and Sentiment by Topic (RQ4)

The reference standard classified 78 of the 209 validation documents as negative, 88 as neutral, and 43 as positive. The transformer model agreed with the reference standard more closely than the other 2 tools on every primary measure (Table 3). Macro-averaged F1 was 0.61 for the transformer model, 0.41 for VADER, and 0.38 for the 2-class DistilBERT model. The tools differed most in how they handled

negative documents. VADER labeled 31 of the 78 negative documents (40%) as positive, against 1 (1%) for the transformer model. Among negative documents, VADER labeled 48% (10/21) of those containing the word positive as positive, compared with 40% (19/47) of those containing neither positive nor negative. On the 5-point scale, the Spearman correlation with the reference standard was 0.55 for the transformer model and 0.42 for VADER. This difference was not statistically significant (difference 0.13, 95% CI -0.02 to 0.28).

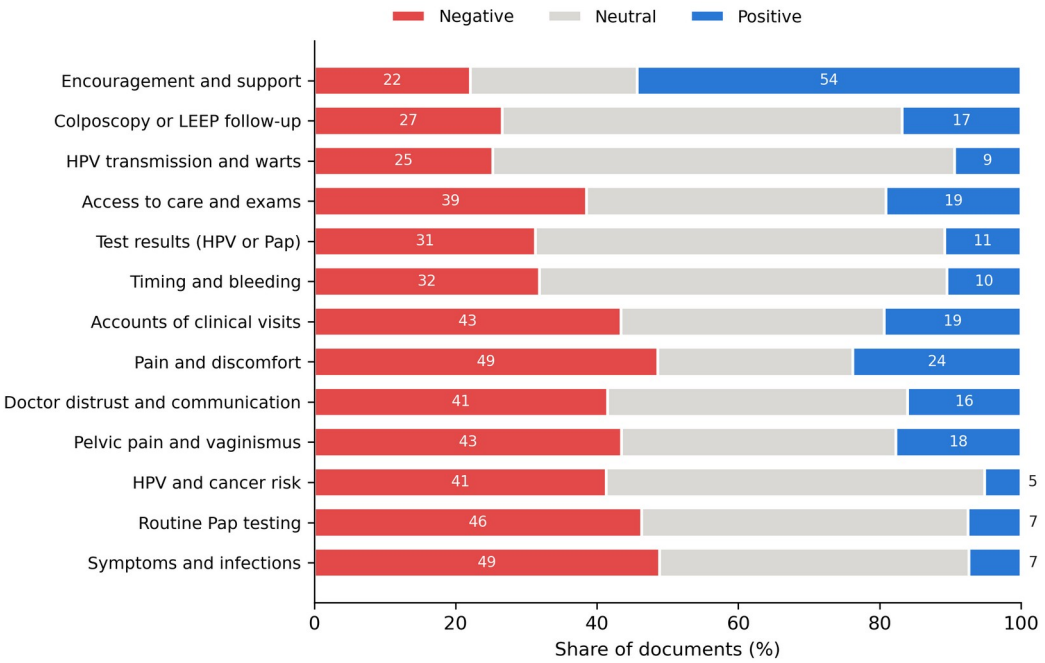
Table 3. Agreement between automated sentiment classification and the human reference standard (n=209 documents).

Tool	Accuracy	Macro F1	Spearman ρ
twitter-roberta-base-sentiment-latest	0.64	0.61	0.55
VADER	0.44	0.41	0.42
distilbert-base-uncased-finetuned-sst-2-english	0.46	0.38	0.52

The reference standard is the consensus of 2 coders. Accuracy and macro F1 compare 3-class labels (negative, neutral, positive) with the reference code collapsed to 3 classes. The DistilBERT model outputs 2 classes and never assigns neutral. Spearman ρ correlates the 5-point reference code with each tool's continuous score. Macro F1 in stratum A (n=130) and stratum B (n=79) was 0.64 and 0.57 for the transformer model, 0.41 and 0.39 for VADER, and 0.35 and 0.42 for DistilBERT. VADER: Valence Aware Dictionary and sEntiment Reasoner.

By the transformer model, 36.8% of documents were negative, 43.6% neutral, and 19.6% positive (Figure 2). Encouragement and support was the only topic in which positive documents outnumbered negative ones (54% vs 22%). Symptoms and infections (49% negative), pain and discomfort (49%), and routine Pap testing (46%) had the largest negative shares, and HPV and cancer risk had the smallest positive share (5%). Sentiment also differed by cluster. The exam-experience cluster had the largest negative share (45.2%), the diagnostic information cluster was predominantly neutral (54.9%), and the narrative and support cluster had the largest positive share (31.1%). The choice of tool changed these conclusions. VADER gave 10 of the 13 topics a positive mean score and the corpus a mildly positive mean (compound 0.15), whereas the transformer model classified more documents as negative than positive in 12 of the 13 topics.

Figure 2. Share of negative, neutral, and positive documents by topic.



Sentiment was classified by twitter-roberta-base-sentiment-latest. Each document is assigned to its highest-loading substantive topic. Numbers are percentages; positive shares below 8% appear to the right of the bar. Topics are ordered by the difference between positive and negative shares.

Fear-Related Language (RQ4)

Of all documents, 13.9% contained at least one explicit fear or anxiety term (Table 1). Fear terms appeared most often in accounts of clinical visits (27.0%), pain and discomfort (21.8%), and test results (18.0%). HPV and cancer risk had the lowest share of all 13 topics (3.5%). Using the full topic mixtures, documents with a fear term loaded more heavily than other documents on accounts of clinical visits (1.9 times the loading of other documents), pain and discomfort (1.5), and timing and bleeding (1.4), and less heavily on HPV and cancer risk (0.4) and HPV transmission (0.4). Of the 2067 sentences containing a fear term, 592 (28.6%) mentioned the procedure or exam, 176 (8.5%) a test result, and 99 (4.8%) the disease; 1335 (64.6%) mentioned none of these. Sentences that mentioned the procedure without the disease outnumbered sentences that mentioned the disease without the procedure by 559 to 66. Adding worry terms did not change the pattern: HPV and cancer risk had the second-lowest share of documents with a term (6.8%), and procedure-only sentences outnumbered disease-only sentences by 698 to 94. With the NRC lexicon, the topic-level result depended on whether clinical terms counted as fear words.

With all NRC fear words, HPV and cancer risk had the highest share of documents with a fear word (75.6%), because cancer and risk are themselves NRC fear words. With clinical terms removed, it again had the lowest share (23.7%). The sentence-level result held in both versions: procedure-only sentences outnumbered disease-only sentences by 2584 to 430 with all NRC fear words and by 1706 to 194 with clinical terms removed.

The network locates these concerns. The 2 topics with the most fear language, accounts of clinical visits and pain and discomfort, had no betweenness and the highest Burt constraint in the network (0.66 and 0.63), so they formed closed neighborhoods. Each connected to 2 of the 3 bridging topics: accounts of clinical visits to timing and bleeding and to pelvic pain and vaginismus, and pain and discomfort to pelvic pain and vaginismus and to access to care. Timing and bleeding, the strongest bridge, was itself among the topics most overrepresented in fear-bearing documents. HPV and cancer risk, the topic with the least fear language, was linked to HPV transmission, test results, and colposcopy or LEEP follow-up within the diagnostic cluster, and its only link to the exam-experience cluster ran through access to care, a topic that fear-bearing documents drew on at half the rate of other documents (0.5). Fear language and negative sentiment did not coincide across topics. Symptoms and infections and HPV and cancer risk were among the most negative topics (49% and 41% negative) yet among the lowest in fear terms (6.2% and 3.5%). Documents whose highest-loading topic fell in the exam-experience cluster made up 29.0% of the corpus and 35.6% of negative documents.

Sensitivity Analysis

Across alpha values from .25 to .35, the backbone ranged from 11 to 44 edges. At alpha=.25 the network fragmented into 5 components and no longer supported a single-network interpretation. Across the connected solutions (alpha=.30 and .35), timing and bleeding remained the highest-betweenness topic, and the separation between diagnostic topics and embodied or experiential topics persisted. The number of clusters varied between 2 and 3, with the 2-cluster solution merging the exam-experience and narrative clusters.

Discussion

Principal Results

Peer discussion of cervical cancer screening centered on the exam, both in how the conversation was structured and in where its fear lay. The discussion divided into three parts: diagnostic information, the embodied experience of the exam, and personal narrative with peer support. Everyday bodily and scheduling concerns held those parts together. Clinical facts occupied more peripheral positions. Talk about bleeding and menstrual timing occupied the most central bridging position, with pelvic pain and access to care close behind. These concerns recur across the whole pathway, since bleeding can be a symptom that prompts a test, a reason to reschedule one, and an expected effect of biopsy or LEEP. Sentiment tracked communicative function. Peer support was the only predominantly positive topic, talk about the exam itself carried the largest share of negative emotion, and diagnostic information was mostly neutral. Fear language followed the same line: it appeared alongside the exam far more often than alongside cancer, and the topic devoted to cancer risk used it least.

Fear Attaches to the Exam More Than to the Disease

The clearest pattern in the emotional texture of this discourse concerns the object of fear. Users rarely used fear terms when discussing HPV and cancer risk, although about 2 in 5 documents on that topic were negative in tone. Fear terms concentrated instead in accounts of clinical visits, pain and discomfort, and test results. Among sentences expressing fear, those that mentioned the procedure without the disease outnumbered the reverse by roughly 7 or 8 to 1. The pattern held with worry terms added and with a crowdsourced emotion lexicon once words naming the disease or the procedure were set aside. It is consistent with reviews that treat fear of the procedure and fear of a cancer diagnosis as separate barriers to screening [3,4,23].

Within the extended parallel process model, the 2 fears play different roles. Fear of cervical cancer is a perceived threat, which can motivate screening when efficacy is high. Because fear of the exam concerns the cost of the protective action itself, we argue that it bears on perceived efficacy. Strong threat combined with low efficacy is the condition under which fear appeals produce defensive responses

[19,20]. Screening differs from many preventive behaviors in this respect, because the recommended action is itself a source of fear. One theoretical implication is that fear appeal research on screening should specify the object of fear, a point an earlier review made for breast cancer screening [23]. Because the analysis rests on words that co-occur in public posts, it describes what users mention alongside fear. It does not measure what individual users fear most or how fear relates to their screening decisions.

The topic network shows how emotion is organized across the conversation, and the pattern sharpens the distinction between the 2 fears. Each cluster carried a distinct emotional signature. The diagnostic information cluster was mostly neutral, and its fear language concentrated in test results and colposcopy follow-up, with almost none in talk about cancer risk and transmission. The exam-experience cluster carried the largest share of negative emotion. The narrative and support cluster combined the most fear-laden topic, accounts of clinical visits, with the only predominantly positive one, encouragement and support, so users who recounted frightening visits and users who offered reassurance met in the same region of the network.

Fear concentrated in enclosed parts of the network. The 2 topics with the most fear language had no bridging role and the most closed neighborhoods, and their connections ran to the bridging topics: timing and bleeding, pelvic pain and vaginismus, and access to care. The bridges therefore sit one step from the places where fear is expressed. Talk about cancer risk, carried with little explicit fear, reached the exam-experience cluster only through access to care. Threat information and procedure fear thus occupy different regions of the conversation, connected by questions about logistics and by concerns about bleeding that recur at every stage of the pathway. Fear and negative sentiment also diverged: the cancer risk and symptom topics were negative in tone with little fear language, so their negativity may reflect other emotions, such as frustration or uncertainty, more than fear. Messages that raise threat therefore address a part of the discussion that carries little explicit fear, and messages about what to expect from the exam address the part where fear is concentrated, which the bridging topics border.

Comparison With Prior Work

The network findings also add to prior work on online health discussion. Previous Reddit health research has shown that these communities serve informational and emotional functions at once [8,9], and computational studies have characterized health discussion on social media [10-12]. This study adds the connective structure. Knowing that encouragement and support is the single most prevalent topic says less, for practical purposes, than knowing that a far less prevalent topic, timing and bleeding, is what links the informational and experiential parts of the conversation. Prevalence and structural position diverged here, and ranking topics by frequency alone would have missed the topics with the widest reach.

The finding also extends the network view of agendas [17,18] to a setting without an agenda setter. An associative topic structure with clusters, bridges, and central nodes emerged from the aggregated contributions of many anonymous users. Because the design is cross-sectional and observational, these results describe the structure of the discourse. They do not establish its effect on what any individual believes.

The sentiment validation carries a methodological caution for social listening in health. A lexicon tool that scores the word positive as positive emotion rated most topics in this discourse as net positive, whereas a transformer model that agreed more closely with human judgment found negative documents outnumbering positive ones in nearly every topic. Lexicon-based sentiment tools should therefore be validated against human coding before use in health contexts, where everyday words often carry clinical meaning.

Implications for Screening Communication

Four implications follow, the first concerning where information should be placed and the next 2 concerning what it should say. First, materials and patient-facing tools that answer concrete bodily and scheduling questions, such as whether to attend an appointment while bleeding, what the exam feels like, and what to do about pelvic pain, address the topics with the widest structural reach in this discourse. The exam-experience topics also carried the largest share of negative emotion, so these materials address the

concerns users find most distressing. Because the bridging topics border the topics where fear concentrates, efficacy content placed alongside answers about bleeding, pelvic pain, and getting an appointment sits next to the conversations in which fear is expressed.

Second, the findings indicate where efficacy messages should point. Messages that stress the severity of cervical cancer address a topic this community discusses with little explicit fear, whereas the fear in this discourse concerned the exam. Within the extended parallel process model [19,20], this suggests that screening messages could direct their efficacy component at the exam itself, and self-collection offers the most direct way to do so. Where it is available, persuasive messages can present self-collection as an option that removes the speculum exam, that people carry out themselves, and that detects precancer as accurately as clinician sampling when polymerase chain reaction-based HPV tests are used [46]. Such messages address both forms of efficacy in the model: confidence that one can complete the test and confidence that the test works. Self-sampling is highly acceptable to women [6], and in trials among underscreened women, mailing self-sampling kits to their homes raised participation compared with invitation letters [46]. For people who undergo a clinician exam, messages can tell readers that they can ask for a smaller speculum, ask the clinician to explain each step or to stop at any point, ask about pain relief, bring a support person, and tell the clinician in advance about pain, vaginismus, or past trauma. Users in this corpus offered one another much of this advice. Messages of this kind build confidence that screening can be less frightening than expected, which speaks to the fear users actually expressed. Future work could test such efficacy messages experimentally against messages that emphasize cancer threat.

Third, programs can make the exam itself less frightening. In trials among women undergoing colposcopy, information leaflets increased knowledge without lowering anxiety, whereas in one trial playing music during the procedure reduced anxiety [47]. Changes in how the exam is delivered can therefore complement messages about it. Survey evidence also shows that practical barriers predict attendance more strongly than emotional ones such as embarrassment [21], which suggests that messages about the exam should accompany practical guidance on getting an appointment. In this network, access

to care was also the topic through which HPV and cancer risk reached the exam-experience cluster, so pairing risk information with guidance on getting and undergoing the exam offers a route from one part of the discussion to the other.

Fourth, for teams building recommendation or content-surfacing features in digital health platforms, topic prevalence is an incomplete guide to placement, and bridging position identifies where accurate information can travel furthest. Platforms could weight bridging topics, such as questions about bleeding and appointments, and the fear-laden topics they border when deciding where to surface accurate screening information.

Limitations

Five limitations qualify these findings. First, the corpus comes from 7 English-language subreddits found through keyword search, and Reddit returns ranked results without retrieving every match. The sample therefore underrepresents discussions that use other vocabulary, and the users of these communities are not representative of all screening-eligible people. Findings describe this discourse. They do not generalize to the screening-eligible population. The availability of self-collection, which informs the implications above, also varies across the countries these users live in. Future work could sample additional platforms and non-English communities, and compare demographic profiles where platform data allow.

Second, topic models are sensitive to preprocessing choices and to k. Coherence guided the choice of k and the author validated the resulting labels, but a different vocabulary cutoff or a different k could shift topic boundaries, most likely by splitting or merging adjacent topics. The broad three-part structure is less likely to change. Comments, which made up most of the corpus, are shorter than submissions and give LDA less information per document, and documents from the same thread are not independent. Future work could compare solutions from several model families on the same corpus and model submissions and comments separately.

Third, agreement between the best-performing sentiment model and the human reference standard was modest (macro F1=0.61), and the reference standard rests on 209 documents. Its errors were concentrated between neutral and the 2 polar classes, and it rarely labeled negative documents as positive. Topic-level sentiment should therefore be read as relative differences between topics; the absolute shares of negative and positive documents are approximate. Future work could fine-tune a sentiment model on a larger set of human-coded health discussion.

Fourth, the fear analysis counted explicit terms. The topic-level results, which show where fear language concentrates, do not depend on identifying what the fear is about. The sentence-level analysis does, and about 2 in 3 fear sentences named none of the objects we tracked; some fear terms are also used figuratively. The NRC check showed that topic-level fear estimates depend on whether clinical terms count as fear words, although the sentence-level pattern held across lexicons. Human coding of the object of fear would test the pattern directly.

Fifth, the network captures co-occurrence of topics across documents and the design is cross-sectional. It does not identify causal structure, show how the discourse formed over time, or demonstrate that reading it changes what users believe or do. Future work could model the network longitudinally around screening campaigns and link exposure to self-reported screening intentions.

Conclusions

Peer discussion of cervical cancer screening centers on the exam. Questions about bleeding, pain, and getting an appointment connect the informational and experiential parts of the conversation and reach further across it than clinical facts do, and the fear users express concerns the exam far more than the disease. Screening communication that answers these everyday questions where they connect the conversation, and that builds confidence about the exam itself, is positioned to reach the widest part of the audience that programs are trying to move.

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Conflicts of Interest

None declared.

Data Availability

The data consist of public Reddit posts and comments collected through the Reddit Data API. The content belongs to the users who posted it, and Reddit's Data API terms do not grant a license to redistribute it, so the corpus is not shared. Account handles were removed before analysis; because public posts can be located through search engines from their text, shared excerpts could still identify users. The analysis code that reproduces every reported result from a Reddit export, and the human sentiment codes for the 209 validation documents (without post text), are available from the corresponding author on reasonable request. The term lists for the fear-language analysis appear in Multimedia Appendix 1.

Authors' Contributions

QP: conceptualization, data curation, formal analysis, investigation, methodology, software, visualization, writing (original draft), and writing (review and editing).

Abbreviations

ANTMN: Analysis of Topic Model Networks
BERT: Bidirectional Encoder Representations from Transformers
HPV: human papillomavirus
LDA: latent Dirichlet allocation
LEEP: loop electrosurgical excision procedure
NRC: National Research Council Canada

RoBERTa: Robustly Optimized BERT Pretraining Approach
VADER: Valence Aware Dictionary and sEntiment Reasoner

Multimedia Appendix 1

Term lists and topic-level results for the fear-language analysis.

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Multimedia Appendix 1. Term lists and topic-level results for the fear-language analysis

Matching was case-insensitive and used word boundaries. Documents were split into sentences at sentence-final punctuation and line breaks.

Fear and anxiety terms

afraid, scared, scary, terrified, terrifying, fear, fears, fearful, frightened, anxious, anxiety, nervous, panic, panicking, panicked, freaking out, freaked out, dread, dreading

Robustness check: worry terms added

worry, worried, worries, worrying

Procedure and exam terms

pap, paps, smear, exam, exams, speculum, colpo, colposcopy, biopsy, biopsies, leep, lletz, procedure, appointment, gyno, gynecologist, obgyn, pelvic, insert, inserted, pain, painful, hurt, hurts, needle, swab

Test result terms

result, results, abnormal, positive, ascus, lsil, hsil, cin, cin1, cin2, cin3

Disease terms (cancer and its consequences)

cancer, cancerous, precancer, precancerous, die, dying, death, tumor, tumour, hysterectomy

Share of documents containing at least one fear or anxiety term, by topic

Topic	Documents, n	With a fear term, %
Accounts of clinical visits	832	27.0
Pain and discomfort	1222	21.8
Test results (HPV or Pap)	862	18.0

Timing and bleeding	201	16.9
Colposcopy or LEEP follow-up	1029	16.7
Pelvic pain and vaginismus	474	15.8
Routine Pap testing	561	14.3
Encouragement and support	1191	12.3
Doctor distrust and communication	1005	9.1
Symptoms and infections	583	6.2
Access to care and exams	592	5.1
HPV transmission and warts	850	4.2
HPV and cancer risk	426	3.5

Across the corpus, 13.9% of documents contained at least one fear or anxiety term. Of 2067 sentences containing such a term, 592 mentioned a procedure or exam term, 176 a test result term, and 99 a disease term; 1335 mentioned none of these. 559 sentences mentioned the procedure without the disease, and 66 mentioned the disease without the procedure. With worry terms added, 17.9% of documents contained a term, the HPV and cancer risk topic had the second-lowest share (6.8%), and 698 sentences mentioned the procedure without the disease against 94 that mentioned the disease without the procedure.

Robustness check with the NRC Emotion Lexicon (fear category, version 0.92)

Version A used all 1474 NRC fear words. Version B removed NRC fear words that name a disease, symptom, body event, or medical procedure: cancer, pain, painful, hurt, hurting, risk, biopsy, procedure,

infection, infectious, medical, bleeding, bloody, diagnosis, birth, surgery, penetration, treat, remove, prevent, hospital, bacteria, disease, shot, cyst, radiation, carcinoma, death, die, dying, tumor, illness, ill, excruciating, malignancy, emergency, cutting, flu, gonorrhea, injection, agony, disorder, swelling, fever, miscarriage, scar, wound, contagious, sting, examination, weight, sweat. Matching used lowercase word tokens. In the sentence-level analysis, procedure and disease terms never counted as fear words.

Topic	Version A, %	Version B, %
Accounts of clinical visits	73.0	51.6
Pain and discomfort	69.6	50.7
Pelvic pain and vaginismus	65.0	44.7
Colposcopy or LEEP follow-up	67.4	41.3
Symptoms and infections	62.4	38.9
Timing and bleeding	59.2	38.8
Encouragement and support	44.7	37.4
Doctor distrust and communication	48.1	36.2
Test results (HPV or Pap)	58.7	35.3
Routine Pap testing	54.7	34.6
HPV transmission and warts	51.2	26.4
Access to care and exams	43.6	26.4
HPV and cancer risk	75.6	23.7

Sentence level: with version A, 2584 sentences mentioned the procedure without the disease and 430 the disease without the procedure; with version B, 1706 and 194.

The NRC Emotion Lexicon is available for non-commercial research from its authors and is not redistributed here.